

discarded amateurish paintings, and such things, the word “attic” would certainly have sprung to mind because each of them has in their memory not only the concept *our own attic* but also the concept *typical attic*, which allows them to envisage a standard attic, if need be. For example, if Kellie were reading a mystery novel and she came across the sentence, “Trembling, the aged aunt slowly groped her way up the steep and narrow stairway towards the attic to look for the golden statuette, but after three quarters of an hour she hadn’t yet come down”, the chance is next to zero that this description would evoke in Kellie’s mind an image of her Boston house’s attic.

This example of the host’s *study*, designated first by his visitors as “your office” and later as “your attic”, shows how we are guided by unconscious analogies towards labels that seem to pinpoint just what we want to say. It illustrates why there is no boundary line — indeed, no distinction — between categorization and the making of analogies.

The Structure of Categories and of Conceptual Space

The anecdote we’ve just related shows that a concept (such as those designated by the terms “attic”, “truck”, “to open”, “to melt”, “to nurse”, “come on!”, “ciao”, and so forth) can have specific and very distinct instances. Indeed, if we ask you to think of a golfer, you *might* conjure up the image of an anonymous middle-aged lady riding, on a Sunday morning, down some anonymous fairway on a golf cart. But it’s more likely that you would conjure up the image of a famous golfer such as Tiger Woods swinging a five-iron, or perhaps you would recall a golf pro you once took lessons from. Instances of the category *golfer* abound, and around each of these specific and concrete instances there is a halo that extends far out. For example, around Tiger Woods, one can imagine seeing him not only making (or missing) a long putt on a tricky green, but also teeing off with great power, hitting out of the rough, and getting out of a sand trap, not to mention appearing in various airport ads and on television, and so forth. Moreover, in this cloud surrounding Tiger Woods, any golf aficionado will surely find a number of Woods’ famous predecessors, such as Jack Nicklaus, Arnold Palmer, Sam Snead, Ben Hogan, and others. Anyone familiar with golf will evoke such images without any trouble at all. And so, what does the concept of *golfer* amount to?

It might seem, *a priori*, that asking about the nature of the concept *golfer* is a minuscule question in comparison with the huge question, “What is human thought all about?”, but the fact is that it is no smaller. In any case, our modest musings about the concept *golfer* are bringing to the fore the obvious fact that concepts are densely stitched together through relationships of similarity and context. The concept of *golfer* is quite closely linked to that of *minigolfer*, and less closely to such concepts as *tennis player*, *runner*, *bicycle racer*, and so forth. Among these connections, some are very close while others are so distant that they barely exist (for example, there will be practically no relation at all between the notions of *golfer* and *sumo wrestler* in anyone’s mind, aside from the fact that both are types of athletes).

The concept of *golfer* is also connected (at various conceptual distances) to a multitude of other concepts, such as *golf course*, *hole*, *fairway*, *tee*, *wood*, *iron*, *putt*, *green*, *par*,

birdie, eagle, bogey, double-bogey, hole-in-one, hook, slice, golf cart, caddie, and tournament, and also, of course, to a large number of specific people (or more precisely, to the concepts one has formed that represent these people). Despite the large number of golfers of whom any fan has certainly heard the name and has very likely stored it in memory, it's far more probable that a fan will think of Tiger Woods than of some middling player from the 1960's. Thus the distance from the "center" of the concept *golfer* to the concept *Tiger Woods* is quite small, whereas the distance from the center out to the middling player from decades ago is very great, unless, perhaps, the particular player happens to be one's mother or one's uncle or something along those lines.

And thus we come to the idea of a multidimensional space in which concepts exist, somewhat like separate points; however, around each such point there is a halo that accounts for the vague, blurry, and flexible quality of the concept, and this halo becomes ever more tenuous as one moves further out from the core.

The Endless Chunking of Concepts in a Human Mind

We could not make an analogy between one concept and another if those concepts had no internal structure in our mind. The very essence of an analogy is that it maps some mental structure onto another mental structure. We can only understand how a hand is analogous to a foot if we mentally recall the fingers and the toes, for instance, as well as the way the hand is physically attached to an arm and the foot to a leg. These kinds of facts are part of what "hand" means; they are integral to the concept *hand*, and they make it what it is. But how many such facts are there "inside" the concept *hand*? How detailed are the internal structures of our concepts? This is the question to which we wish to turn for a moment.

Consider a rather complex memory in the mind of a certain professor — say, the sabbatical year she spent in Aix-en-Provence. When she recalls that year, of course she doesn't replay its 300-plus days like a movie; rather, she sees just the tiniest part of it, in its barest outlines. It's as if she were looking down at a mountain range from an airplane, but an extensive cloud layer allowed only a handful of the chain's highest peaks to peek through.

If someone asks her about details of the city of Aix, or about some major event that happened during the year, or about the most interesting people she met there, or about the schools that her children went to, and so on, then any of these aspects will become available upon request, but until that happens, they are all hidden under the "cloud cover". And if she decides to shift her focus to the school that her children attended that year, then still just a handful of the school's most salient aspects will come into view. If her focus shifts still further down onto a particular teacher, then a handful of that person's most salient features come into view — and on it goes. The overarching memory — the sabbatical year in Aix — is never seen in its full glory; rather, just a tiny (but very salient) fraction of it is ever made available. However, pieces of it can be focused in on, and in this way, the large memory can be unpacked into its component pieces, and the same can be done to those pieces, in turn.

Perhaps they will also think, “Most of Frontier’s flights go in and out of Denver”, or else “Lots of Frontier planes and gates are found at the Denver airport”. This small set of “highest peaks” (*i.e.*, most salient facts) is pretty much all that one needs in most cases where a hub is being talked about. But in fact it leaves out nearly all of what makes up the concept *hub*, and of which most adults in our culture are perfectly aware. The stipulation “in our culture” is crucial, because hundreds of concepts that we take for granted are not part of other cultures or eras. For instance, imagine trying to explain the concept of *hub* to Johann Sebastian Bach, or to Joan of Arc, or to Archimedes, or to Nebuchadnezzar. These were all remarkable individuals in their respective cultures — but how would you go about trying to get across this “simple” concept to any one of them? It would be a rather long story.

To begin with, the word “hub” is the name of a very concrete, visual concept that we learn when we first ride a bicycle and we see the many spokes radiating out of (or into) the wheel’s center — its hub. Indeed, it’s because of wheels with spokes that airline hubs sport the name “hub”, and the concept of *bicycle wheel* is certainly more “primitive” or “elementary” than is that of *airline hub*, not only being learned far earlier in life but also being far simpler to grasp. Let’s list some other concepts that are more primitive than *hub* and that are likewise prerequisites to it. There’s *airline*, for instance, and *route* and *schedule* and *route map*. And in order to understand the concept of *airline*, you first have to be familiar with the concepts of *airplane* and *company*. And the concept of *route* depends on the concepts of *starting point*, *destination*, *leg*, and *connection*. We won’t go on forever, but let’s not forget that the *raison d’être* of hubs is economic efficiency — the relentless pressure to cut costs and to reduce the number of different flights — and thus one has to know about the concepts of *trade*, *gain*, *loss*, *profits*, *competition*...

We have only scratched the surface of what goes into the concept *hub*. All of those ingredients are “in there”, and they could all, if and when the need arose, be unpacked and revealed. Such unpackings carry one back towards more and more basic, elementary notions — concepts involving motion, vehicles, roundness, acquisition, trading, winning and losing, large and small numbers, and on and on. And note that none of what we have so far spoken of has anything to do with the fact that airports are associated with large cities, or that airports are more than just black dots on maps — indeed, we completely skipped the internal *physical* structure of an airport, with its runways, tarmacs, concourses, gates, jetways, food courts, etc.

The image we’ve just given of a chain of concepts that depend on other concepts, moving ever downwards in complexity, is reminiscent of the nesting of Russian dolls, and might give the impression that concepts are in fact structured in this boxlike fashion. In truth, however, the phenomenon of concept-building is much subtler and more fluid than that. Concepts are not like nested boxes, with any given concept being rigidly defined in terms of a precise set of previously-acquired concepts, and with concepts always being acquired in a fixed order. Instead, when new concepts are acquired, their arrival often exerts a major impact upon the “more primitive” concepts on which they are based, a bit as if the construction of a house affected the very nature of the bricks with which it was built. Although houses that modify the nature of the

bricks of which they are made do not exactly grow on trees, we are all nonetheless familiar with this basic idea, since, for example, children are dependent upon parents in order to exist, but at the same time their existence radically transforms the lives of their parents.

This is true also for concepts. Thus the concept of *hub* depends, without any doubt, on many others, such as *airport*, but at the same time, the concept of *airport* is itself modified by the concept of *hub*. For instance, familiarity with the *hub* idea inevitably brings out the fact that airports are entities that can help airlines to become more streamlined and thus to save money; this notion is certainly not the most obvious fact about airports. Similarly, recalling that airports tend to be transit areas for travelers somewhat reduces the saliency of airports as final destinations. Even if such effects do not cause radical modifications of the concept *airport*, they are undeniably real and demonstrate that the original concept doesn't remain unaffected by the newer one. One can imagine more radical effects of the *hub* concept on that of *airport*, such as novel kinds of architecture aimed at optimizing the design of airports to function as hubs, or the design of new kinds of airport shopping malls specially designed to serve passengers who are making rapid plane-changes and who have only twenty or thirty minutes. And the existence of hubs can change the seemingly obvious correlation between city size and airport size; that is, with hubs, it becomes perfectly conceivable for a relatively small city (such as Charlotte, North Carolina) to have an airport with an enormous volume of air traffic but very few passengers who actually disembark there. We thus see that although there cannot be a "child" concept of *hub* without the prior "parent" concept of *airport*, the child nonetheless changes the identity of the parent.

There are countless examples of this general sort. It happens particularly often in science, where a new idea depends intrinsically on previous ones, but at the same time it casts the old ones in a fresh new light, and often a deeper light. For example, non-Euclidean geometry not only came historically out of Euclidean geometry, but it also allowed a much deeper understanding of Euclidean geometry to emerge. In physics, much the same could be said for relativistic mechanics and quantum mechanics, both of which are "children" of classical mechanics, and together have yielded a far deeper understanding of it.

The same is true for concepts in everyday life. Thus, the relatively new notions of *surrogate mother*, *adoptive mother*, and *single mother* all come out of the concept of *mother*, as does that of a homosexual couple that adopt a child, and each of these new notions modifies the concept of *mother*, showing how a mother need not give birth to a child, need not raise a child, need not be part of a couple, and may even not be a female. In like manner, the concept of *divorce* depends on that of *marriage*, and yet it also has reverse effects on the nature of *marriage* itself (think, for instance, of the effect of prenuptial contracts, and of the fact that today everyone knows, when going into a marriage, that half of all marriages finish in divorce). The notion of *homosexual marriage* clearly depends on the prior concept of *marriage*, and the intensity of the debate over homosexual marriage is in large part due to the fact that opponents claim that the idea not only extends the concept of *marriage* but in fact does the concept serious harm. The concept